15. Write the python program to implement Decision Tree

Aim: To implement a Decision Tree classifier using Python and classify data based on its features.

Procedure:

1. Import the required libraries.
2. Create a sample dataset with input features and class labels.
3. Separate the features and target labels.
4. Train a Decision Tree classifier.
5. Predict the class for a new input.
6. Display the predicted result.

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```
from sklearn.tree import DecisionTreeClassifier

X = [
    [1, 1],
    [1, 0],
    [0, 1],
    [0, 0],
    [1, 1],
    [0, 0]
]

y = ['Yes', 'Yes', 'Yes', 'No', 'Yes', 'No']

model = DecisionTreeClassifier(random_state=0)
model.fit(X, y)

prediction = model.predict([[1, 0]])

print("Predicted Class:", prediction[0])
```

Output:

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```
Python 3.14.7 (tags/v3.14.7
AMD64)] on win32
Enter "help" below or click
>>>
========== RESTART: C:
Predicted Class: Yes
>>>
```

Result: Thus, a Decision Tree classifier was implemented successfully using Python.